

Abhinav Prakash

abhinavprakash616@gmail.com — +91 94575 48199 — Noida, India — [linkedin.com/in/abhipra24](https://www.linkedin.com/in/abhipra24) — Portfolio —
GitHub Pages

SUMMARY

- SDET with **5 years** of experience owning the architecture of highly scalable test automation frameworks, CI/CD pipelines, and shared engineering utilities **built from scratch** for complex hardware-software ecosystems – including automotive-adjacent EV charging systems communicating over OCPP 1.6/2.0.1.
- Strong programming expertise in Python and Java, with hands-on ownership of BDD (Behave)-driven and API-level test automation, transforming manual QA processes into fully automated, CI/CD-integrated pipelines across the SDLC.
- Deep expertise in DevSecOps and network monitoring (IIT Roorkee PGP), integrating tools like Burp Suite and Wireshark directly into automated pipelines to ensure secure, reliable execution for consumer-facing hardware products.

EXPERIENCE

Senior Software QA Engineer

Apr 2025 – Present

ChargePoint, Gurugram, India

- Served as technical project QA lead, driving end-to-end test infrastructure architecture and automation strategy for critical EV charging (hardware-software) releases – full technical ownership without direct people management.
- Architected "DevUtils" and "ARM" (Automation Resource Management) from the ground up: Python, FastAPI, and Streamlit-based shared utilities for system setup, deep log analysis, and API-level test orchestration, reducing defect triage time by 30%.
- Built and scaled modular BDD automation frameworks (Python/Behave) and integrated them into Jenkins CI/CD pipelines using Docker, achieving **40% faster execution** across deployment cycles.
- Delivered results validating high-reliability embedded systems logic – power-sharing algorithms and secure OTA firmware updates over OCPP – mirroring the hardware-software sync rigor required for automotive/vehicle systems.

Software QA Engineer

Apr 2022 – Apr 2025

ChargePoint, Gurugram, India

- Took ownership as first-line debugger for developer teams, writing custom Python and Shell scripts to continuously parse Unix system logs and query RDBMS tables, cutting defect triage time by 30%.
- Designed scalable BDD test automation frameworks from scratch using Python, Behave, and Playwright, **reducing manual infrastructure setup and test effort by 25%**.
- Leveraged cybersecurity expertise to embed security checks into automated runs, using Wireshark and Burp Suite to monitor network traffic and validate API security.
- Drove comprehensive backend and API-level testing of high-priority REST workflows, ensuring fault-tolerant communication between edge devices and cloud services.

Programmer Analyst Trainee

Feb 2021 – Mar 2022

Cognizant

- Automated critical UI test cases for CVS Health projects using Selenium with Java, **improving pipeline reliability and test coverage**.
- Contributed to framework enhancements and execution pipelines, supporting Agile delivery cycles and improving overall CI efficiency.

EDUCATION

PGP, Cyber Security

Jul 2023 – Mar 2024

Indian Institute of Technology, Roorkee

B.Tech, Electronics & Communications Engineering

2017 – 2021

Jaypee Institute Of Information Technology

SKILLS

Frameworks & Programming: Python, Java, Behave (BDD), Pytest, Playwright, Selenium, FastAPI, Streamlit

DevOps & Infrastructure: Unix, Shell Scripting, Linux, Jenkins, Docker, CI/CD Pipelines, Scalable Test Framework Architecture

Databases & Security: RDBMS (SQL/PostgreSQL), Wireshark, Burp Suite, Network Monitoring

Protocols & APIs: REST API Testing (API-level automation), Postman, Swagger, OCPP 1.6/2.0.1 (Hardware-Software Integration)

CERTIFICATIONS

Learning Jenkins — 5 Common Test Failures — Data Structures and Algorithms – Coding Blocks